

Topic 13: File Handling in C++

12.1 Why File Handling?

- RAM is temporary — all program data is lost when the program ends
- Files on disk are permanent — data persists between runs
- C++ uses file streams from `<fstream>` header
- Three stream types: `ofstream` (write), `ifstream` (read), `fstream` (both)

12.2 Writing to a File

```
#include <fstream>
ofstream file("data.txt");
if (file.is_open()) {
    file << "Ahmad" << endl;
    file << "Marks: 85" << endl;
    file.close();
}
```

- `ofstream` creates or overwrites the file
- `file.is_open()` checks if the file opened successfully — always check
- `file.close()` flushes and closes the file

12.3 Reading from a File

```
ifstream file("data.txt");
string line;
while (getline(file, line)) {
    cout << line << endl;
}
file.close();
```

- `getline(file, line)` reads one line at a time
- The while loop runs until the file ends (getline returns false at EOF)

12.4 File Open Modes

- `ios::out` — write, overwrite existing (default for `ofstream`)
- `ios::app` — append to end without deleting existing content

- `ios::in` — read (default for ifstream)
- Combined: `fstream file("data.txt", ios::in | ios::out);`
- `ofstream file("data.txt", ios::app);` — append mode

12.5 File Handling Flowchart

- Open file → check if open → read or write in loop → close file
 - Always close files to ensure data is written and resources are freed
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